



**SYDNEY BOYS HIGH SCHOOL**  
MOORE PARK, SURRY HILLS

**2006**

YEAR 12  
TRIAL HIGHER SCHOOL  
CERTIFICATE

# Mathematics Extension 1

## General Instructions

- Working time – 2 Hours
- Reading time – 5 Minutes
- Write using black or blue pen
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may not be awarded for messy or badly arranged work.
- Hand in your answer booklets in 4 sections. Section A (Questions 1 and 2), Section B (Questions 3 and 4), Section C (Questions 5 and 6) and Section D (Question 7).

## Total Marks – 84

- Attempt Questions 1 – 7.
- All QUESTIONS are of equal value.

Examiner: *R. Boros*

**Section A – Start a new booklet****Marks****Question 1. (12 marks)**

- a)
- (i) Evaluate  $\int_0^1 \frac{x}{x^2+1} dx$  leaving your answer in exact form. 2
- (ii) Evaluate  $\int_{-2}^{2\sqrt{3}} \frac{1}{x^2+4} dx$  leaving your answer in exact form. 2
- b) Find the gradient of the tangent to the curve  $y = \tan^{-1}(\sin x)$  at  $x = 0$ . 2
- c) Solve for  $x$ ,  $\frac{1}{x+1} < 3$ . 2
- d) Give the general solution of the equation,  $\cos\left(\theta + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$ . 2
- e) If  $f(x) = 8x^3$ , then find the inverse function  $f^{-1}(x)$ . 2

**Question 2. (12 marks)**

- a) Find the co-ordinates of the point  $P$  that divides the interval  $A(-4, -6)$  and  $B(6, -1)$  externally in the ratio 3:1. 2
- b)
- (i) Sketch the graph of  $y = |2x - 4|$ . 2
- (ii) Using your graph, or otherwise, solve the inequation  $|2x - 4| > x$ . 2
- c) Use the substitution  $u = 1 + x$  to evaluate,  $\int_{-1}^3 x\sqrt{1+x} .dx$ . 2
- d) Solve for  $n$ ,  $2 \times {}^nC_4 = 5 \times {}^nC_2$ . 2
- e) What is the least distance between the circle  $x^2 + y^2 + 2x + 4y = 1$  and the line  $3x + 4y = 6$ ? (Leave your answer in exact form.) 2

**End of Section A**

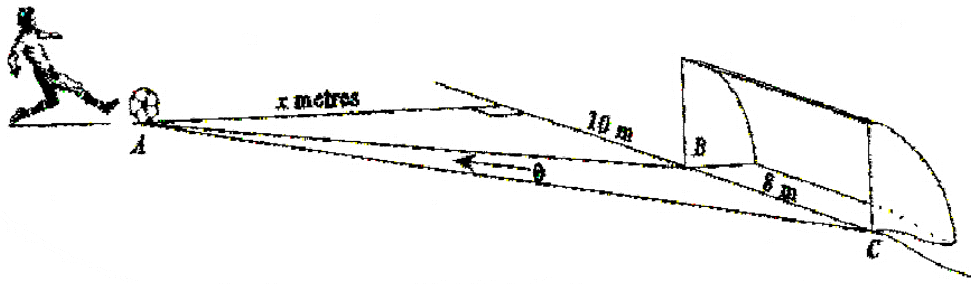
**Section B – Start a new booklet****Marks****Question 3. (12 marks)**

- a) If the roots of the equation,  $x^4 - 2x^3 - 5x + 1 = 0$ , are  $t_1, t_2, t_3, t_4$ ,  
find  $\sum_{i=1}^4 (t_i t_j t_k)^{-1}$ , such that  $i \neq j \neq k$ . 2
- b) State the domain and range of the function  $y = 2 \sin^{-1}\left(\frac{x}{3}\right)$ .  
Hence sketch the curve. 3
- c) A bowl of water heated to  $100^\circ\text{C}$  is placed in a coolroom where the temperature is maintained at  $-5^\circ\text{C}$ . After  $t$  minutes, the temperature  $T^\circ\text{C}$  of the water is changing so that  $\frac{dT}{dt} = -k(T + 5)$ .
- (i) Prove that  $T = Ae^{-kt} - 5$  satisfies this equation and find the value of  $A$ . 1
- (ii) After 20 minutes, the temperature of the water has fallen to  $40^\circ\text{C}$ . How long, to the nearest minute, will the water need to be in the coolroom before ice begins to form, (i.e. the temperature falls to  $0^\circ\text{C}$ ). 2
- d) (i) Show that the equation  $\ln x + x^2 - 4x = 0$  has a root lying between  $x = 3$  and  $x = 4$ . 2
- (ii) By taking  $x = 4$  as a first approximation, use one application of Newton's Method to obtain another approximation for the root, to 2 decimal places. Is this newer approximation a better one? Explain. 2

**Question 4. (12 marks)****Marks**

- a)** The points  $P(2ap, ap^2)$  and  $Q(2aq, aq^2)$  lie on the parabola  $x^2 = 4ay$ . It is given that the chord  $PQ$  has equation  $y = \left(\frac{p+q}{2}\right)x - apq$ .
- (i) Show that the gradient of the tangent at  $P$  is  $p$ . 1
- (ii) Prove that if  $PQ$  passes through the focus, then the tangent at  $P$  is parallel to the normal at  $Q$ . 2
- b)** A committee of five is to be formed from 4 Liberal senators, 3 Labor senators and 2 Democrat senators.
- (i) How many different committees can be formed that have 3 Liberals, 1 Labor and 1 Democrat? 1
- (ii) If the committee is to be chosen at random, what is the probability that there will be a Liberal majority in the committee? 2
- c)**
- (i) Express  $7\cos\theta - \sin\theta$  in the form  $R\cos(\theta + \alpha)$ , where  $R > 0$  and  $0^\circ \leq \alpha \leq 90^\circ$ . 2
- (ii) Hence solve  $7\cos\theta - \sin\theta = 5$  for  $0^\circ \leq \theta \leq 360^\circ$ , giving your answer to the nearest degree. 2
- d)** Find the values of the constants  $a$  and  $b$  if  $x^2 - 2x - 3$  is a factor of the polynomial  $P(x) = x^3 - 3x^2 + ax + b$ . 2

**End of Section B**

**Section C – Start a new booklet****Marks****Question 5. (12 marks)****a)**

A soccer player A is  $x$  metres from a goal line of a soccer field. He takes a shot at the goal  $BC$ , with the ball not leaving the ground.

- (i) Show that the angle  $\theta$  within which he must shoot is given by

$$\theta = \tan^{-1} \left( \frac{8x}{180 + x^2} \right) \text{ when he is 10 metres to one side of the near}$$

goal post and 18 metres to the same side of the far post.

2

- (ii) Find the value of  $x$  which makes this angle a maximum. (Leave your answer in exact form).

2

- b)** A particle moves in a straight line such that its velocity  $V$  m/s is given by

$V = 2\sqrt{2x-1}$  when it is  $x$  metres from the origin. If  $x = \frac{1}{2}$  when  $t = 0$  find:

- (i) the acceleration.

1

- (ii) an expression for  $x$  in terms of  $t$ .

2

**c)**

Find the volume of the solid obtained by rotating  $y = \sin^{-1} x$  about the  $y$ -axis

between  $y = -\frac{\pi}{4}$  and  $y = \frac{\pi}{4}$ . Answer in exact form.

3

**d)**

The perimeter of a circle is increasing at 3 cm/s. Leaving your answer in terms of  $\pi$ , find the rate at which the area is increasing when the perimeter is 1m.

2

**Question 6. (12 marks)****Marks****a)**

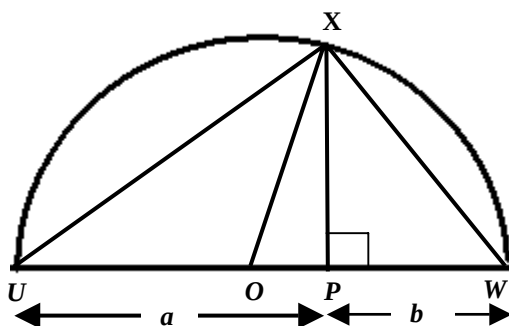
Consider the following three expressions involving  $n$ , where  $n$  is a positive integer:

$$5^n + 3, 7^n + 5, 5^n + 7$$

- (i) By substituting values of  $n$ , show that  $7^n + 5$  is the only one of these expressions which could be divisible by 6 for all positive integers  $n$ . 1
- (ii) Use mathematical induction to show that the expression  $7^n + 5$  is in fact divisible by 6 for all positive integers  $n$ . 2

**b)**

Not to scale



In the diagram  $UXW$  is a semi-circle with  $O$  as a midpoint of diameter  $UW$ . The point  $P$  lies on  $UW$  and  $XP$  is perpendicular to  $UW$ . The length of  $UP = a$  units and  $PW = b$  units are shown.

- (i) Explain why  $OX = \frac{a+b}{2}$ . 1
- (ii) Show that  $\angle UXP \equiv \angle XWP$ . 1
- (iii) Deduce that  $XP = \sqrt{ab}$ . 1
- (iv) By using the diagram show that  $\frac{a+b}{2} \geq \sqrt{ab}$ . 1

**c)**

The displacement  $x$  metres of a particle from the origin is given by

$$x = 5 \cos\left(3t - \frac{\pi}{6}\right), \text{ where } t \text{ is the time lapsed in seconds.}$$

- (i) Show that  $\ddot{x} = -9x$ . 1
- (ii) Find the period of the motion 1

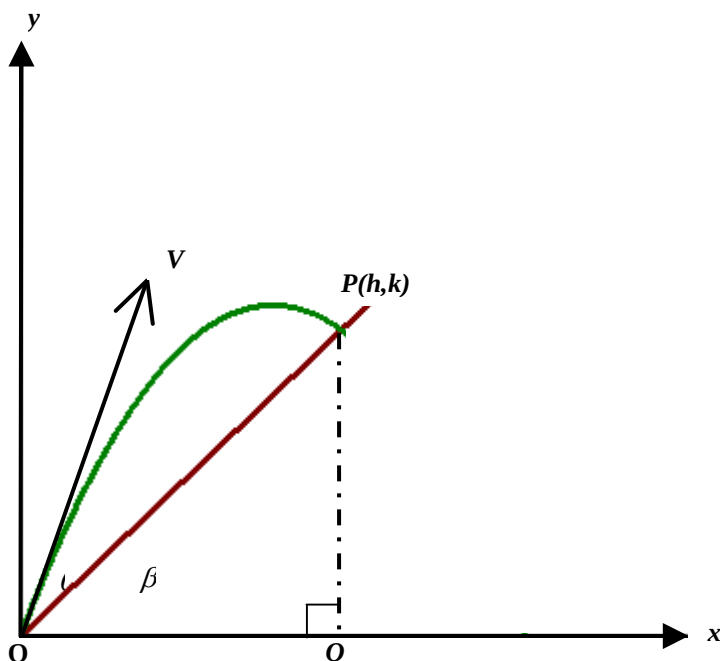
**Marks**

**d)** Suppose that  $(5 + 2x)^{12} = \sum_{k=0}^{12} a_k x^k$ .

(i) Use the binomial theorem to write the expression for  $a_k$ . 1

(ii) Show that  $\frac{a_{k+1}}{a_k} = \frac{24 - 2k}{5k + 5}$  2

**End of Section C**

**Section D – Start a new booklet****Marks****Question 7. (12 marks)**

A projectile is fired from the origin with a velocity  $V$  and an angle of elevation  $\theta$ , where  $\theta \neq 90^\circ$ . You may assume that  $x = Vt \cos \theta$  and  $y = -\frac{1}{2}gt^2 + Vt \sin \theta$ , where  $x$  and  $y$  are the horizontal and vertical displacements of the projectile in metres from  $O$  at time  $t$  seconds after firing, and  $g$  is the acceleration due to gravity.

- (i) Show that the Cartesian equation of the flight of the projectile is:

$$y = x \tan \theta - \frac{g}{2V^2 \cos^2 \theta} x^2 \quad 1$$

- (ii) Suppose the projectile is fired up a plane inclined at  $\beta$  to the horizontal so that  $0^\circ \leq \beta \leq \theta$ . If the projectile strikes the plane at  $P(h, k)$ , show that:

$$h = \frac{(\tan \theta - \tan \beta) 2V^2 \cos^2 \theta}{g} \quad 2$$

- (iii) Hence, show that the range  $OP$  of the projectile can be given by

$$OP = \frac{2V^2 \sin(\theta - \beta) \cos \theta}{g \cos^2 \beta} \quad 4$$

**Marks**

- (iv) Given the fact that  $2 \sin(x - \beta) \cos x = \sin(2x - \beta) - \sin \beta$ . Show that the maximum value of the range of  $OP$  is given by:

$$\frac{V^2}{g(1 + \sin \beta)} \quad 4$$

- (v) If the angle of inclination of the plane is  $14^\circ$ , at what angle to the horizontal should the projectile be fired in order to attain maximum range? 1

**End of Examination**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE:  $\ln x = \log_e x, x > 0$

Question 1:

$$\begin{aligned} \text{a) i)} \int_0^1 \frac{x}{x^2+1} dx &= \frac{1}{2} \int_0^1 \frac{2x}{x^2+1} dx \quad \frac{1}{2} \\ &= \frac{1}{2} \left[ \ln(x^2+1) \right]_0^1 \quad \frac{1}{2} \\ &= \frac{1}{2} [\ln 2 - \ln 1] \\ &= \frac{1}{2} \ln 2 \quad 1 \end{aligned}$$

$$\begin{aligned} \text{ii)} \int_{-2}^{2\sqrt{3}} \frac{1}{x^2+4} dx &= \left[ \frac{1}{2} \tan^{-1} \frac{x}{2} \right]_{-2}^{2\sqrt{3}} \quad \frac{1}{2} \\ &= \left( \frac{1}{2} \tan^{-1} \frac{2\sqrt{3}}{2} \right) - \left( \frac{1}{2} \tan^{-1} \frac{-2}{2} \right) \\ &= \left( \frac{1}{2} \tan^{-1} \sqrt{3} \right) - \left( \frac{1}{2} \tan^{-1} (-1) \right) \\ &= \left( \frac{1}{2} \cdot \frac{\pi}{3} \right) - \left( \frac{1}{2} \cdot \frac{-\pi}{4} \right) \quad \frac{1}{2} \\ &= \frac{\pi}{6} - - \frac{\pi}{8} \\ &= \frac{7\pi}{24} \quad 1 \end{aligned}$$

b) Find the gradient of the tangent to the curve  $y = \tan^{-1}(\sin x)$  at  $x = 0$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{1 + (\sin x)^2} \times (\cos x) \quad -1 \text{ for no } x \cos x. \\ &= \frac{\cos x}{1 + \sin^2 x} \quad 1 \end{aligned}$$

when  $x = 0$

$$\begin{aligned} \frac{dy}{dx} &= \frac{\cos 0}{1 + \sin^2 0} \\ &= \frac{1}{1} \end{aligned}$$

$$= 1$$

c) solve for  $x$ ,  $\frac{1}{x+1} < 3$

$$x + 1 \neq 0$$

$$\therefore x \neq -1 \quad \frac{1}{2}$$

$$\times \text{ by } (x+1)^2$$

$$\frac{(x+1)^2}{x+1} < 3(x+1)^2 \quad \frac{1}{2}$$

$$x+1 < 3x^2 + 6x + 3$$

$$0 < 3x^2 + 6x + 3 - (x+1)$$

$$0 < 3x^2 + 5x + 2$$

$$\frac{(3x+3)(3x+2)}{3}$$

$$0 < (x+1)(3x+2) \quad \frac{1}{2}$$

$$= \frac{3(x+1)(3x+2)}{3}$$

$$x < -1, \quad x > -2/3 \quad \frac{1}{2}$$

d) General solution for

$$\cos(\theta + \pi/4) = 1/\sqrt{2}$$

$$\cos(\theta + \pi/4) = \cos \pi/4 \quad \frac{1}{2}$$

$$\theta + \pi/4 = 2n\pi \pm \pi/4 \quad 1$$

$$\theta = 2n\pi \pm \pi/4 - \pi/4 \quad \frac{1}{2}$$

e)  $f(x) = 8x^3$  find inverse function  $f^{-1}(x)$

$$f(f^{-1}(x)) = x = f^{-1}(f(x)) \quad \frac{1}{2}$$

$$f(f^{-1}(x)) = x$$

$$\therefore 8(f^{-1}(x))^3 = x$$

$$f^{-1}(f(x)) = \sqrt[3]{8x^3}$$

$$f^{-1}(x)^3 = x/8$$

$$\therefore f^{-1}(x) = \sqrt[3]{\frac{x}{8}}$$

$$= \frac{2x}{2}$$

$$= \frac{\sqrt[3]{x}}{2} \quad \frac{1}{2}$$

$$= x \quad \frac{1}{2}$$

~~$$f^{-1}(x) = \frac{\sqrt[3]{8x^3}}{2}$$~~

$$\therefore f^{-1}(x) = \frac{\sqrt[3]{8x^3}}{2} \quad \frac{1}{2}$$

~~$$f^{-1}(x) = \frac{\sqrt[3]{8x^3}}{2}$$~~

$$-1 \text{ for } \sqrt[3]{8} = 2\sqrt{2}$$

$$-1 \text{ for } \frac{\sqrt[3]{x}}{8}$$

## Question 2.

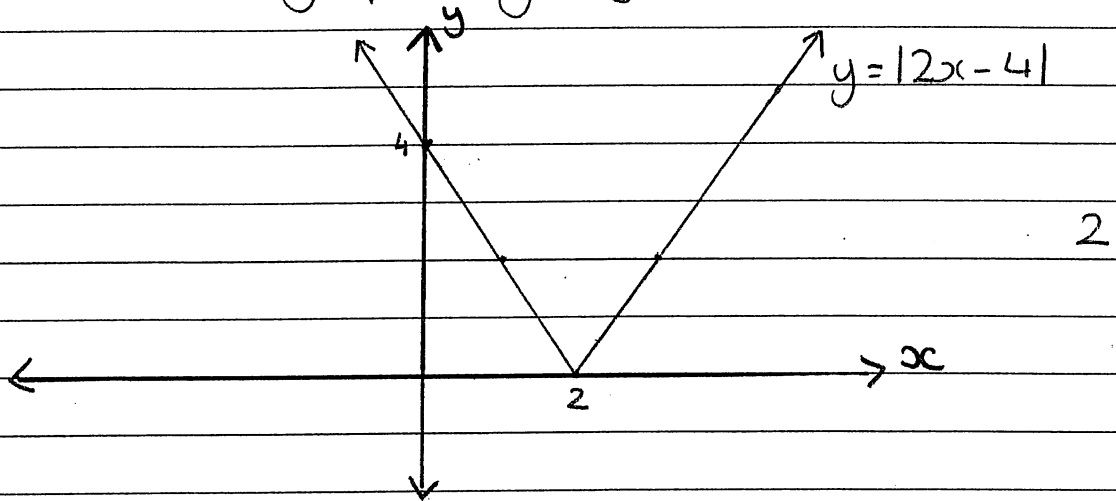
a)  $A(-4, -6)$   $B(6, -1)$  divide externally in ratio  $3:1$

$$x\text{-co-ordinate} = \frac{(3 \times 6) - (1 \times -4)}{3 - 1} = 11 \frac{1}{2}$$

$$y\text{-co-ordinate} = \frac{(3 \times -1) - (1 \times -6)}{3 - 1} = \frac{3}{2} \frac{1}{2}$$

$\therefore$  The co-ordinates of P are  $(11, \frac{3}{2})$    
  $\frac{1}{2}$  if  $x_2, x_1, y_2, y_1$  are switched.

b) i) sketch the graph of  $y = |2x - 4|$



ii) solve  $|2x - 4| > x$

$$\sqrt{(2x - 4)^2} > x$$

$$(2x - 4)^2 > x^2$$

$$4x^2 - 16x + 16 > x^2$$

$$3x^2 - 16x + 16 > 0$$

$$\frac{(3x - 12)(3x - 4)}{3} > 0$$

$$\frac{3(x - 4)(3x - 4)}{3} > 0$$

$$x > 4, x < \frac{4}{3} \quad 2$$

c) Use  $u = 1+x$  to evaluate  $\int_{-1}^3 x \sqrt{1+x} dx$

$$u = 1+x$$

limits:

$$\therefore \frac{du}{dx} = 1 \quad \therefore x = u-1 \quad \begin{array}{l} x=3 \therefore u=1+3 \Rightarrow u=4 \\ x=-1 \therefore u=1-1 \Rightarrow u=0 \end{array}$$

$$du = dx$$

$$\int_0^4 (u-1) \cdot \sqrt{u} du \quad 1/2$$

$$= \int_0^4 (u-1) \cdot u^{1/2} du$$

$$= \int_0^4 (u^{3/2} - u^{1/2}) du$$

$$= \left[ \frac{u^{5/2}}{5/2} - \frac{u^{3/2}}{3/2} \right]_0^4 \quad 1/2$$

$$= \left[ \frac{2 \cdot 4^{5/2}}{5} - \frac{2 \cdot 4^{3/2}}{3} \right] - \left[ \frac{2 \cdot 0^{5/2}}{5} - \frac{2 \cdot 0^{3/2}}{3} \right]$$

$$= \left[ \frac{2 \times 32}{5} - \frac{2 \times 8}{3} \right]$$

$$= \frac{64}{5} - \frac{16}{3}$$

$$= \frac{112}{15} \text{ OR } 7\frac{7}{15} \quad 1/2$$

d) solve for  $n$ ,  $2 \times {}^nC_4 = 5 \times {}^nC_2$   $-1/2$  for  $n=-3$

$$\frac{2 \times n!}{(n-4)! 4!} = \frac{5 \times n!}{(n-2)! \times 2!}$$

$$\div \text{ by } n! \quad \frac{2 \cdot 1}{24 (n-4)!} = \frac{5 \cdot 1}{2 (n-2)!}$$

$$\times \text{ by } (n-4)! \quad \frac{2}{24} = \frac{5}{2} \cdot \frac{1}{(n-2)(n-3)}$$

$$\times \text{ by } 2/5 \quad \frac{1}{30} = \frac{1}{(n-2)(n-3)}$$

check:

$$2 \times {}^8C_4 = 140 = 5 \times {}^8C_2$$

$$\therefore n = 8. \quad 1$$

✓

2e) circle  $x^2 + y^2 + 2x + 4y = 1$

$$\therefore (x+1)^2 + (y+2)^2 = 6$$

$$\therefore \text{centre } (-1, -2) \quad \text{radius } \sqrt{6} \quad \frac{1}{2}$$

$$\text{line } 3x + 4y = 6 \Rightarrow 3x + 4y - 6 = 0$$

least distance between circle & line is  
the distance between the line & centre of the circle less the radius.

$$d = \frac{|(3 \times -1) + (4 \times -2) - 6|}{\sqrt{3^2 + 4^2}}$$

$$= \frac{|-3 - 8 - 6|}{5}$$

$$= \frac{|-17|}{5} \quad \frac{1}{2}$$

$$\therefore \text{minimum distance} = \frac{17}{5} - \sqrt{6} \quad \frac{1}{2}$$

17/5 1 mark only.

# SECTION B QUESTION 3

$$a) \sum_{\substack{i=1 \\ i \neq j \neq k}}^4 (t_i t_j t_k)^{-1} = \frac{1}{t_1 t_2 t_3} + \frac{1}{t_1 t_2 t_4} + \frac{1}{t_1 t_3 t_4} + \frac{1}{t_2 t_3 t_4}$$

$$= \frac{t_1 + t_2 + t_3 + t_4}{t_1 t_2 t_3 t_4}$$

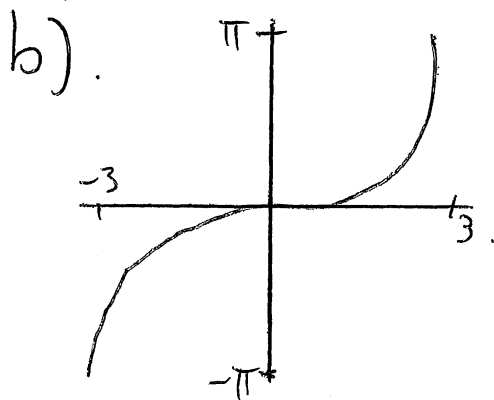
$$t_1 + t_2 + t_3 + t_4 = -\frac{b}{a}$$

$$= 2$$

$$t_1 t_2 t_3 t_4 = \frac{c}{a}$$

$$= 1$$

$$\text{So } \sum_{\substack{i=1 \\ i \neq j \neq k}}^4 (t_i t_j t_k)^{-1} = 2$$



Domain:  $-1 \leq \frac{x}{3} \leq 1$   
 $-3 \leq x \leq 3$

Range:  $-\frac{\pi}{2} \leq \sin^{-1}\left(\frac{x}{3}\right) \leq \frac{\pi}{2}$

$$-\pi \leq 2 \sin^{-1}\left(\frac{x}{3}\right) \leq \pi$$

$$-\pi \leq y \leq \pi$$

$$c i) \quad LHS = \frac{dT}{dt}$$

$$= -kAe^{-kt}$$

$$RHS = -k(T+S)$$

$$\begin{aligned} &= -k(Ae^{-kt} - 5 + 5) \\ &= -kAe^{-kt} \\ &= LHS. \end{aligned}$$

Initial conditions

$$100 = Ae^{-k0} - 5$$

$$A = 105.$$

ii) After 20 minutes

$$40 = 105e^{-20k} - 5$$

$$\frac{45}{105} = e^{-20k}$$

$$-20k = \ln \frac{3}{7}$$

$$k = \frac{\ln \frac{3}{7}}{-20}.$$

At  $0^\circ C$ .

$$0 = 105e^{-kt} - 5$$

$$e^{-kt} = \frac{5}{105}$$

$$t = \frac{\ln \frac{1}{21}}{-k}.$$

$$t = 72 \text{ minutes.}$$

di) Since  $f(x) = \ln x + x^2 - 4x$  is a continuous function and

$$f(3) = \ln 3 + 3^2 - 4 \times 3 \\ \approx -1.9 < 0$$

and

$$f(4) = \ln 4 + 4^2 - 4 \times 4 \\ \approx 1.4 > 0$$

Therefore  $f(x) = \ln x + x^2 - 4x$  must have a root between  $x=3, x=4$ .

$$ii) f'(x) = \frac{1}{x} + 2x - 4.$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

$$x_0 = 4$$

$$x_1 = 4 - \frac{\ln 4 + 16 - 16}{\frac{1}{4} + 8 - 4}$$

$$\approx 3.67.$$

Yes, since we know  $f(x)$  has a root between 3 and 4 and this approximation is closer to 3 than the first approximation of 4.

### QUESTION 4.

i).  $y = \frac{x^2}{4a}$

$$\frac{dy}{dx} = \frac{x}{2a}.$$

At  $P(2ap, ap^2)$

$$m_p = \frac{2ap}{2a} \\ = p.$$

ii) Similarly to part (i) the tangent at Q is  $m_q = q$ .

Thus the gradient of the normal will be  $m = -\frac{1}{q}$ .

Given that the chord goes through the Locus  $(0, a)$ . then

$$a = \left(\frac{p+q}{2}\right)0 - apq.$$

$$pq = -1$$

$$\therefore q = -\frac{1}{p}.$$

Thus the gradient of the normal at Q will be  $-\frac{1}{q} = -\left(-\frac{1}{p}\right)$

$$= p.$$

∴ Tangent at P is parallel to the normal at Q.

$$b). i) \binom{4}{3} \binom{3}{1} \binom{2}{1} = 24.$$

$$ii) n(\text{Sample space}) = \binom{9}{5} = 126.$$

$$\begin{aligned} n(E) &= n(4 \text{ l.b}) + n(3 \text{ l.b}) \\ &= \binom{4}{4} \binom{5}{1} + \binom{4}{3} \binom{5}{2} \\ &= 5 + 40 \\ &= 45. \end{aligned}$$

$$\begin{aligned} P(E) &= \frac{n(E)}{n(S)} \\ &= \frac{45}{126} \\ &= \frac{5}{14}. \end{aligned}$$

$$c) \quad R = \frac{\sqrt{49+1}}{5\sqrt{2}}$$

$$\tan \alpha = \frac{1}{7}$$

$$\alpha = 8^{\circ} 8'$$

$$\text{So } 7\cos\theta - \sin\theta = 5\sqrt{2} \cos(\theta + 8^{\circ} 8').$$

$$ii) 5\sqrt{2} \cos(\theta + 88^\circ) = 5$$

$$\cos(\theta + 88^\circ) = \frac{1}{\sqrt{2}}$$

$$\theta + 88^\circ = 45^\circ, 315^\circ$$

$$\theta \approx 37^\circ, 307^\circ$$

$$d) p(-1) = (-1)^3 - 3(-1)^2 + a(-1) + b$$

$$= -4 - a + b$$

$$p(3) = (3)^3 - 3(3)^2 + (3)a + b$$

$$= 3a + b$$

$$\text{So } -4 - a + b = 0$$

$$\text{and } 3a + b = 0$$

Ⓐ  
Ⓑ

$$\text{from } \textcircled{A} \quad a = b - 4$$

sub into Ⓑ

$$3b - 12 + b = 0$$

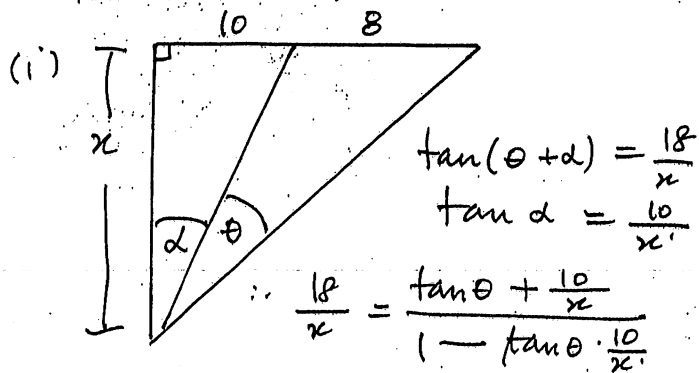
$$b = 3$$

$$a = -1$$

So

$$\therefore a = -1, b = 3$$

# Question (5)



$$(\tan \theta)x^2 - 8x + 180 \tan \theta = 0$$

$$\tan \theta = \frac{8x}{180 + x^2}$$

$$\therefore \theta = \tan^{-1} \left( \frac{8x}{180 + x^2} \right)$$

$$\frac{d\theta}{dx} = \frac{(180 + x^2)8 - 8x(2x)}{(x^2 + 180)^2}$$

$$1. \quad \frac{1440 - 8x^2}{(x^2 + 180)^2} = 0$$

$$8(x^2 - 180) = 0, \quad x^2 = 180$$

$$x = 6\sqrt{5}$$

Test:

|                      |    |             |       |
|----------------------|----|-------------|-------|
| $\theta$             | 13 | $6\sqrt{5}$ | 14    |
| $\frac{d\theta}{dx}$ | +  | 0           | -1568 |

(1)

(b)  $v = 2(2x - 1)^{\frac{1}{2}}$

(i)  $v^2/2 = 4x - 2$

$$\dot{x} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right) = 4$$

[1]  
(1)

(ii)  $\frac{dx}{dt} = 2(2x - 1)^{\frac{1}{2}}$

$$\frac{dx}{2x - 1} = \frac{1}{2} dt$$

$$\therefore t = \frac{1}{2} (2x - 1)^{\frac{1}{2}} + C$$

When  $t = 0$ ,  $x = \frac{1}{2} \Rightarrow C = 0$

$$\therefore 2t = \sqrt{2x - 1}$$

$$1 + 4t^2 = 2x$$

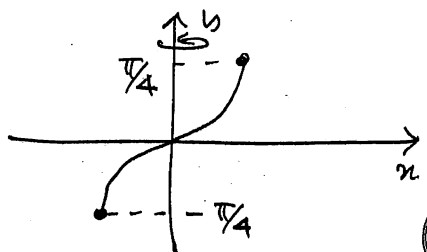
$$\therefore x = \frac{1 + 4t^2}{2}$$

$$= 2t^2 + \frac{1}{2}$$

(2)

[2]

(c)



$$\left( \sin^2 y = \frac{1 - \cos 2y}{2} \right)$$

$$V = 2\pi \int_0^{\pi/4} \sin^2 y \, dy$$

$$= \pi \int_0^{\pi/4} (1 - \cos 2y) \, dy$$

$$= \pi \left[ y - \frac{\sin 2y}{2} \right]_0^{\pi/4}$$

$$= \pi \left( \frac{\pi}{4} - \frac{1}{2} \right) \quad [3]$$

(d)

$$\frac{dp}{dt} = 3, \quad p = 2\pi r$$

$$\therefore t = \frac{p}{2\pi}$$

$$A = \pi \left( \frac{p^2}{4\pi^2} \right) = \frac{p^2}{4\pi}$$

$$\frac{dA}{dt} = \frac{dA}{dp} \cdot \frac{dp}{dt} = \frac{p}{2\pi} \times 3$$

$$p = 100$$

$$\therefore \frac{dA}{dt} = \frac{150}{\pi} \text{ cm}^2/\text{s}$$

$$= \frac{0.015}{\pi} \text{ m}^2/\text{s} \quad [2]$$

(12)

(2)

## Question (6)

6(a)  $n = 1, 6 \nmid 8$

(i)  $7 + 5 = 12$  [1]

(ii) For  $n = 1$ ,  $7 + 5 = 12$  and  $6 \mid 12$  (1)

Assume  $S(k)$  is true

i.e.  $7^5 + 5 = 6M$ ,  $M \in \mathbb{N}$

Consider  $n = k + 1$

$$7^{k+1} + 5$$

$$= 7 \cdot 7^k + 5$$

$$= 7(7^5 + 5) - 30$$

$$= 42M - 30 = 6(7M - 5)$$

$$= 6N \text{ where } N = 7M - 5 \in \mathbb{N}$$

$\therefore S(k+1)$  is true when  $S(k)$

is true and  $S(1)$  is true

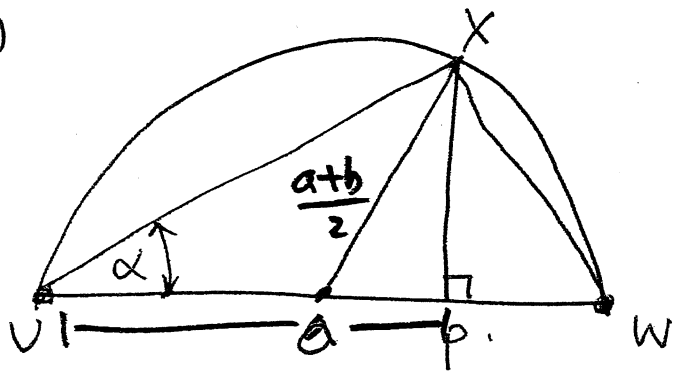
$\therefore$  by mathematical induction

$$6 \mid (7^n + 5) \quad \forall n \in \mathbb{N}$$

(2)

[2]

(b)



$OX = \frac{1}{2} UW = a + b$ .  
(radius =  $\frac{1}{2}$  diameter)

$$\therefore OX = \frac{a+b}{2} \quad \text{--- (1)} \quad [1]$$

(ii) Let  $\angle XUP = \alpha$ .

$\angle UXW = 90^\circ$  (Angle in a semicircle)

$$\therefore \angle XWP = 90^\circ - \alpha.$$

In  $\triangle UXP$ ,  $\triangle XWP$ .

$$\angle XWP = 90 - \alpha \Rightarrow \angle XWP = \alpha$$

(Angle sum of  $\triangle XPW$ .)

$$\therefore \triangle UXP \parallel \triangle XWP \quad [1]$$

(equiangular)

In  $\triangle XPU$

$$XP^2 + a^2 = XU^2$$

$$\frac{XP}{PW} = \frac{UP}{XP}$$

$$\therefore XP^2 = a \cdot b$$

In  $\triangle XWP$

$$XP^2 + b^2 = XW^2.$$

In  $\triangle UXW$

$$(a+b)^2 = XU^2 + XW^2$$

$$a^2 + b^2 + 2ab = (a^2 + b^2) + 2XP^2$$

$$\therefore XP^2 = ab \quad [1]$$

$$\Rightarrow XP = \sqrt{ab}.$$

In  $\triangle OXP$   $OX \perp XP$   
( $OX$  is the hypotenuse)  $[1]$

$$\therefore \frac{a+b}{2} > \sqrt{ab}.$$

$$(c) \quad \ddot{x} = -15 \sin\left(3t - \frac{\pi}{6}\right)$$

$$\begin{aligned} (i) \quad \ddot{x} &= -45 \cos\left(3t - \frac{\pi}{6}\right) \\ &= -9 \left[ 5 \cos\left(3t - \frac{\pi}{6}\right) \right] \\ &= -9x \end{aligned} \quad [1]$$

$$(ii) \quad T = \frac{2\pi}{\omega} = \frac{2\pi}{3}. \quad [1]$$

$$(d) (i) \quad a_k = \binom{12}{k} 5^{12-k} 2^k. \quad [1]$$

$$(ii) \quad \frac{a_{k+1}}{a_k} = \frac{\binom{12}{k+1} 5^{11-k} 2^{k+1}}{\binom{12}{k} 5^{12-k} 2^k} = \frac{2}{5} \frac{(12-k)}{k+1} \quad [2]$$

# QUESTION 7

$$(i) t = \frac{x}{v \cos \theta}$$

$$y = -\frac{gx^2}{2v^2 \cos^2 \theta} + \frac{vx \sin \theta}{v \cos \theta}$$

$$y = x \tan \theta - \frac{gx^2}{2v^2 \cos^2 \theta}$$

(ii)

$$\text{At P, } y = k = h \tan \theta, x = h$$

$$h \tan \beta = h \tan \theta - \frac{gh^2}{2v^2 \cos^2 \theta} \text{ from (i)}$$

$$\frac{gh^2}{2v^2 \cos^2 \theta} = h(\tan \theta - \tan \beta)$$

$$h = \frac{(\tan \theta - \tan \beta) 2v^2 \cos^2 \theta}{g}$$

(iii)

$$OP = \frac{h}{\cos \beta}$$

$$= \frac{(\tan \theta - \tan \beta) 2v^2 \cos^2 \theta}{g \cos \beta} \text{ [from (ii)]}$$

$$= \frac{\left( \frac{\sin \theta}{\cos \theta} - \frac{\sin \beta}{\cos \beta} \right) 2v^2 \cos^2 \theta}{g \cos \beta}$$

$$= \frac{(\sin \theta \cos \beta - \sin \beta \cos \theta) 2v^2 \cos \theta}{g \cos^2 \beta}$$

$$= \frac{2v^2 \sin(\theta - \beta) \cos \theta}{g \cos^2 \beta}$$

(iv)

$$OP = \frac{[\sin(2\theta - \beta) - \sin \beta] v^2}{g \cos^2 \beta} \text{ (given)}$$

$$\frac{d(OP)}{(d\theta)} = \frac{2v^2}{g \cos^2 \beta} [2 \cos(2\theta - \beta)]$$

$$OP \text{ max/min } \cos(2\theta - \beta) = 0$$

$$2\theta - \beta = 90^\circ$$

$$\theta = \frac{90^\circ + \beta}{2}$$

$$OP'' = \frac{4v^2}{g \cos^2 \beta} \times -2 \sin(2\theta - \beta)$$

$$\text{always} < 0 \text{ as } (2\theta - \beta) < 180^\circ$$

$$\therefore \text{max val OP when } \theta = \frac{90^\circ + \beta}{2}$$

$$\text{Max val. OP} = \frac{v^2 (\sin 90^\circ - \sin \beta)}{g(1 - \sin^2 \beta)}$$

$$= \frac{v^2 (1 - \sin \beta)}{g(1 - \sin^2 \beta)}$$

$$= \frac{v^2}{g(1 + \sin \beta)}$$

(v)

$$\text{max val OP when } \theta = \frac{90^\circ + \beta}{2} \text{ [from (iv)]}$$

$$\theta = \frac{90^\circ + 14^\circ}{2}$$

$$\theta = 52^\circ$$